

EXPERIMENT-13

13. Implement your own ping program.

ping_program.py

```
import socket
import time

host = input("Enter host/IP address: ")

try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")

    for i in range(4):
        start = time.time()

        try:
            s = socket.create_connection((ip, 80), timeout=2)
        except:
            print("Request timed out.")
            time.sleep(1)

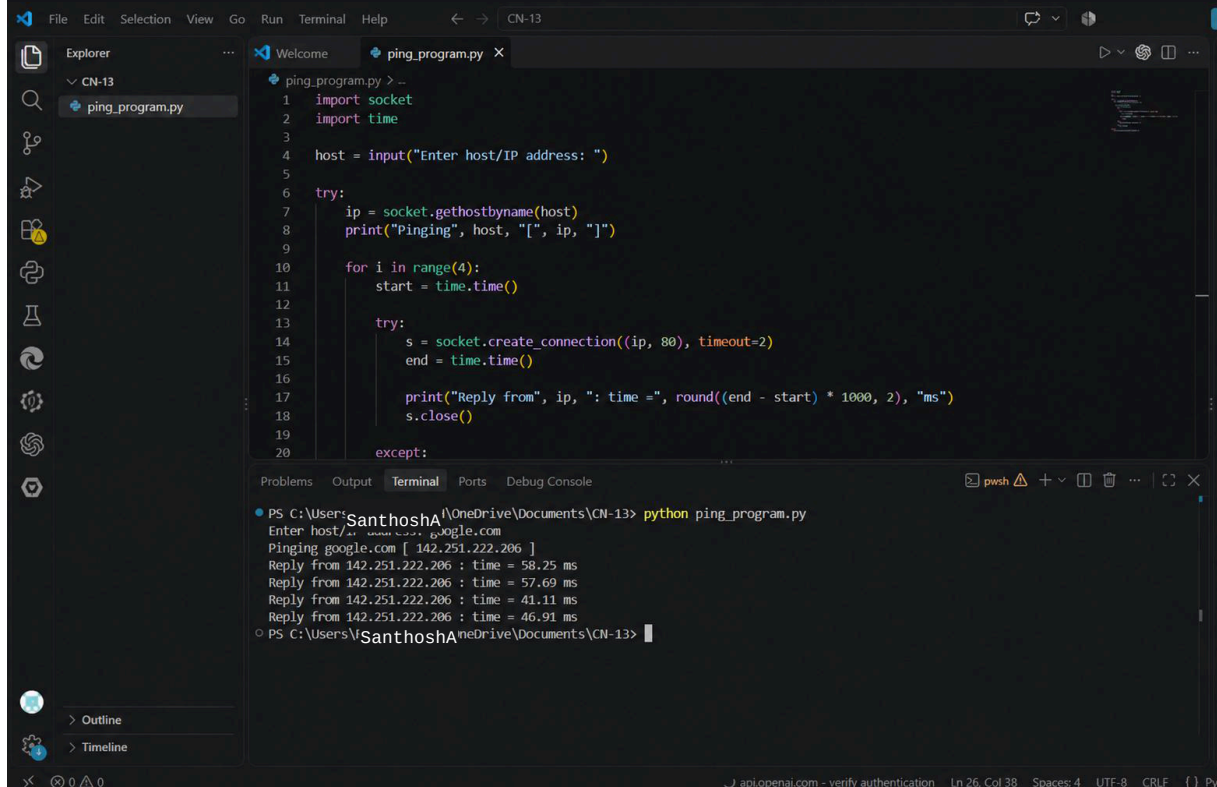
        end = time.time()

        print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")

    s.close()

except:
    print("Invalid host/IP address.")
```

```
PS C:\Users\Santhosha\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\Santhosha\OneDrive\Documents\CN-13>
```



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a file named `ping_program.py` under a folder named `CN-13`. The main editor window displays the code for `ping_program.py`. The code imports `socket` and `time`, prompts the user for a host/IP address, and then performs a ping test using `socket.create_connection` with a timeout of 2 seconds. It prints the IP address and the response time for each of four attempts.

```
1 import socket
2 import time
3
4 host = input("Enter host/IP address: ")
5
6 try:
7     ip = socket.gethostbyname(host)
8     print("Pinging", host, "[", ip, "]")
9
10    for i in range(4):
11        start = time.time()
12
13        try:
14            s = socket.create_connection((ip, 80), timeout=2)
15            end = time.time()
16
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
18            s.close()
19
20 except:
```

Below the code editor, the Terminal panel shows the output of running the script. It displays the same command prompt output as shown in the first block, including the host address and the four ping response times.

```
PS C:\Users\Santhosha\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\Santhosha\OneDrive\Documents\CN-13>
```